

NONCOOPERATIVE GAME THEORY — ECE270: HOMEWORK #5

Exercise 11 (Transformed games). Consider a zero-sum matrix game defined by an $n \times m$ matrix

$$A = \begin{bmatrix} a_{11} & a_{12} & \cdots & a_{1m} \\ a_{21} & a_{22} & \cdots & a_{2m} \\ \vdots & \vdots & \ddots & \vdots \\ a_{n1} & a_{n2} & \cdots & a_{nm} \end{bmatrix}$$

1. Consider a second zero-sum matrix game defined by the following $n \times m$ matrix

$$B = \begin{bmatrix} a_{11} + \beta & a_{12} + \beta & \cdots & a_{1m} + \beta \\ a_{21} + \beta & a_{22} + \beta & \cdots & a_{2m} + \beta \\ \vdots & \vdots & \ddots & \vdots \\ a_{n1} + \beta & a_{n2} + \beta & \cdots & a_{nm} + \beta \end{bmatrix},$$

where $\beta \in \mathbb{R}$ is a constant. In essence, outcomes of this second game are obtained from those of the original game by adding the constant β to the original outcome.

Show that a pair of mixed policies (y^*, z^*) is a mixed saddle-policies equilibrium for A if and only if it is a mixed saddle-policies equilibrium for B .

2. Consider now a third zero-sum matrix game defined by the following $n \times m$ matrix

$$C = \begin{bmatrix} f(a_{11}) & f(a_{12}) & \cdots & f(a_{1m}) \\ f(a_{21}) & f(a_{22}) & \cdots & f(a_{2m}) \\ \vdots & \vdots & \ddots & \vdots \\ f(a_{n1}) & f(a_{n2}) & \cdots & f(a_{nm}) \end{bmatrix}, \quad (4)$$

where $f : \mathbb{R} \rightarrow \mathbb{R}$ is a monotone increasing function, in the sense that

$$x \geq y \Rightarrow f(x) \geq f(y), \quad \forall x, y \in \mathbb{R}.$$

Show that if (i^*, j^*) is a pure saddle-policies equilibrium for A then (i^*, j^*) is also a pure saddle-policies equilibrium for C .

3. Assume that the function f in (4) is monotone strictly increasing, in the sense that

$$x > y \Rightarrow f(x) > f(y), \quad \forall x, y \in \mathbb{R}.$$

Show that a pair of policies (i^*, j^*) is a pure saddle-policies equilibrium for A if and only if it is a pure saddle-policies equilibrium for C .

Hint: In contrast with the previous problem, you now also need to show that if (i^, j^*) is a pure saddle-policies equilibrium for C then (i^*, j^*) is also a pure saddle-policies equilibrium for A .*

Notation. Games with the same set of saddle-point equilibria are called *best-response equivalent*.

4. Assume that the function f in (4) is a monotone strictly increasing affine function, i.e., that

$$f(x) = \alpha x + \beta, \quad \forall x \in \mathbb{R},$$

for some $\beta \in \mathbb{R}$ and $\alpha > 0$. Show that a pair of policies (y^*, z^*) is a mixed saddle-policies equilibrium for A if and only if it is a mixed saddle-policies equilibrium for C . $\square$

Exercise 12. Show that for any bimatrix game, the mixed security level for a given player P_i is always smaller (i.e., better) than or equal to the pure security level for the same player P_i . $\square$